

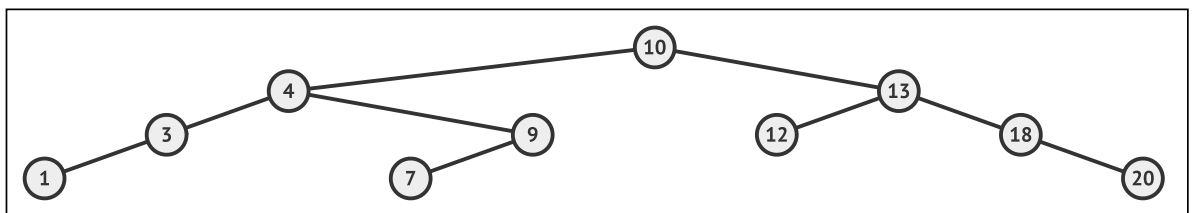
Course: DSA

Midterm Examination

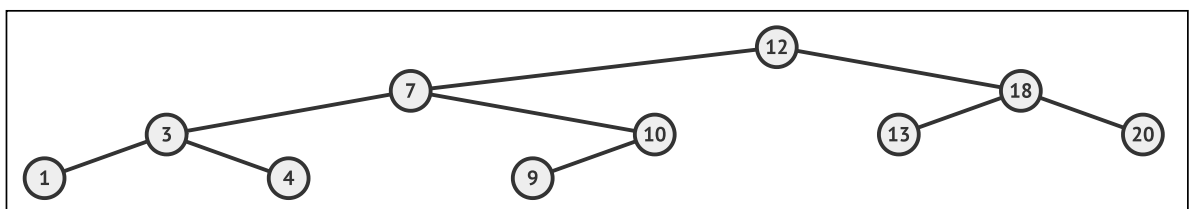
June 18, 2024

- Q1.** [1pt] Given a set of keys $\{1, 2, 3, 4, 5\}$, list all the permutations where their inversion numbers are 7
- Q2.** [2pt] Draw a BST that has the PreOrder output $\{79, 24, 4, 3, 61, 39, 56, 41, 59, 71, 67, 84, 94, 91\}$
- Q3.** [2pt] Given a set of keys $\{1, 2, 3, 4, 5\}$, draw all the BSTs where their heights are 4
- Q4.** [2pt] Given the two following BSTs, using rotation operations to transform the first BST to the second BST, draw and describe each step

BST 1



BST 2



Q5. [2pt] Implement DSW algorithm and demonstrate how to use it

Q6. [1pt] Clear writing

Note

- Save your work as a PDF file named “midterm-[id].pdf”.
- Submit the file through Moodle